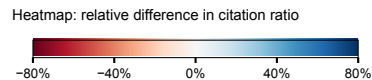
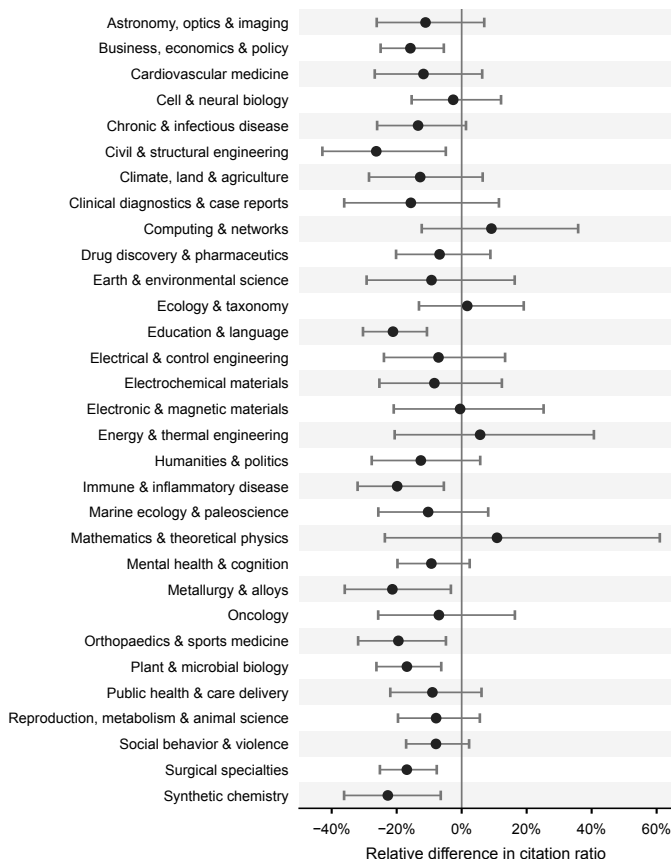


a Across research areas

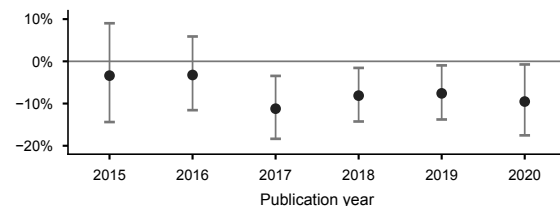


27 of 31 areas and all six years
had lower estimated ratios.

b Within each area, year by year

	2015	2016	2017	2018	2019	2020
Astronomy, optics & imaging	-37	-7	-13	-8	-3	-6
Business, economics & policy	-9	-17	-10	-11	-22	-19
Cardiovascular medicine	+1	+6	-33	+1	+21	-25
Cell & neural biology	-7	-3	-3	+5	0	-5
Chronic & infectious disease	-2	+9	-13	-3	-10	-18
Civil & structural engineering	-25	-9	-21	-23	-36	-29
Climate, land & agriculture	+19	-20	-15	-21	-19	-9
Clinical diagnostics & case reports	+6	+25	-3	-10	-4	-47
Computing & networks	+74	-9	+1	+5	+7	+3
Drug discovery & pharmaceuticals	-1	+6	-24	-15	-17	+7
Earth & environmental science	-6	+8	-39	+10	-13	-4
Ecology & taxonomy	+6	+21	-2	-18	+19	-3
Education & language	-27	-11	-28	-23	-29	-11
Electrical & control engineering	+27	-8	-15	-16	-15	+2
Electrochemical materials	0	+74	-3	-18	-26	-15
Electronic & magnetic materials	-16	-22	+13	+34	+1	+6
Energy & thermal engineering	-28	-12	-11	+27	+23	+19
Humanities & politics	-37	-18	-11	-4	-3	-13
Immune & inflammatory disease	x	x	x	x	x	-27
Marine ecology & paleoscience	x	x	x	x	x	x
Mathematics & theoretical physics	+14	+41	+7	+19	0	-6
Mental health & cognition	-11	-8	-13	-7	+1	-10
Metallurgy & alloys	x	x	-34	-31	-10	+8
Oncology	-15	-10	+3	-13	-5	-2
Orthopaedics & sports medicine	x	x	x	-22	-29*	-10
Plant & microbial biology	0	-18	-24	-21	-10	-21
Public health & care delivery	-24	-14	-1	+1	0	-8
Reproduction, metabolism & animal science	x	x	x	x	x	x
Social behavior & violence	-2	-18	-11	-5	-6	-7
Surgical specialties	-18	-24	-16	-19	-4	-20
Synthetic chemistry	x	x	x	x	x	x

c Across publication years



Ratio: other-area / same-topic citations. Negative values: lower in narrower-scope journals.